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Walking Speed and Uncertainty Estimation Using Mixture Density Networks for Dynamic Ambulatory Environments

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***Abstract*— Walking speed, often considered a representative indicator of activity levels, becomes notably reduced as muscle strength and cardiovascular function decline with aging. Wearable walking rehabilitation devices aim to alleviate the effort during walking or enhance the necessary muscles. Measuring the wearer's walking speed provides an objective assessment of rehabilitation progress. While various methods, such as GPS, model-based estimation, and deep neural network regression can estimate walking speed, they encounter challenges in diverse environments. This article introduces the CNN-based Mixture Density Network (CMDN) structure, which enhances accuracy and provides uncertainty information about estimated walking speed, indirectly reflecting the current walking environment. Validated with experiments involving 20 elderly individuals, CMDN demonstrated performance across flat and stair descent situations, showcasing its potential as a foundation for widespread use in diverse scenarios.**

***Keywords*—Walking speed estimation, convolutional neural network, mixture density network, human activity recognition**

I. Introduction

As people age, our physical health becomes increasingly important, and walking speed serves as a valuable metric in assessing the well-being of the elderly [1], as walking speed is greatly affected by muscular strength and cardiovascular function which are major health indicators. The gradual decline in walking speed is often accompanied by a reduction in overall physical activity [2]. In this regard, the monitoring of walking speed and interventions to enhance it emerges as a simple yet powerful strategy for daily rehabilitation [3], [4]..

Recently developed walking rehabilitation wearable devices include various sensors that recognize walking status and estimate walking speed [5], [6]. Based on the measured data, these devices monitor the wearer's health status and assist in returning to a healthy life [7], [8]. Sensors used in walking rehabilitation wearable devices are broadly divided into three categories: GPS, inertial measurement unit (IMU) and rotary encoder, all of which can be used for estimating walking speed. The advantage of the GPS method is its accuracy [9]. However, it is vulnerable to signal interference from terrain or building structures and is not usable indoors. Methods using rotary encoder or IMU to estimate walking speed include model-based estimation and deep neural network (DNN) based estimation. In model-based estimation, the inverted pendulum model is often used [10], utilizing the acceleration in the direction of gravity. While the method provides explanatory power, precise control over the attachment positions of IMU and rotary encoder is required, along with additional personal information such as the patient's height or leg length. Rotary encoders, especially, depends on the alignment with the center of leg joints [11], leading to varying reliability based on the wearing state of the wearable device, posing a risk of not working robustly in devices using various torque application methods [12]. In the case of using DNN, it shows higher accuracy than model-based methods [5], but it requires high computational power and lacks explanatory power for situations not covered by the training data. Acquiring training data for everyday walking situations with diverse environments such as slopes and stairs is challenging [13].

Mixture Density Network (MDN) employs multiple Gaussian distributions to represent a given probability distribution [14]. When integrated with Deep Neural Network (DNN) regression, MDN can enhance performance without a significant increase in computational demand [15]. Especially, MDN facilitates the quantification of uncertainty in the outputs of regression tasks. Uncertainty is categorized into aleatoric and epistemic types [16], where aleatoric uncertainty denotes the inherent uncertainty present in the training data, while epistemic uncertainty refers to the uncertainty that can be mitigated by acquiring additional training data. By distinguishing between these types of uncertainties, MDN enhances the interpretability of model predictions, indicating whether a prediction is influenced by inherent noise or by conditions not represented in the training environment.

In this article, we incorporated a Mixture Density Network (MDN) into a Deep Neural Network (DNN) to estimate walking speed, aiming to evaluate two primary effects. Firstly, the MDN was expected to provide improved accuracy compared to the conventional DNN. This is particularly relevant when implementing DNNs on wearable devices with limited embedded memory, which restricts the expansion of the network's depth and the number of parameters. Secondly,

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by calculating the uncertainty of the output data, it becomes possible to discern whether the walking environment is similar to the flat surfaces, which predominate in the training dataset, or presents more dynamic scenarios, such as slopes or stairs, which are typically underrepresented in training data. With the limited scope of training environments, the calculated uncertainty allows us to selectively trust the prediction output of the model. We employed a walking speed dataset acquired on a flat surface for training, while we assessed the model's uncertainty evaluation performance using a dataset from stair descents.

II. Methods

A. Walking Speed Dataset in Real World

The walking speed dataset used in this article was obtained from 20 elderly individuals (average age: 75.0 ± 5.1, average height: 1.57 ± 0.07 m, average weight: 60.2 ± 12.1 kg, 4 males). The Participants walked outdoors for approximately 20 minutes wearing wearable devices (Fig. 1) and acquired sensor values at 20 Hz through the built-in IMU and the rotary encoder. Simultaneously, walking speed was measured at 1 Hz using the GPS of a smartphone (SM-A336N, Samsung Electronics). Subsequently, the GPS data underwent a moving average filter (window size 3) and was synchronized with the IMU and the rotary encoder data through linear interpolation. Stair descent data was collected in the same manner for use in uncertainty estimation. Human subject-based experiment is conducted with the experimental procedures approved by the

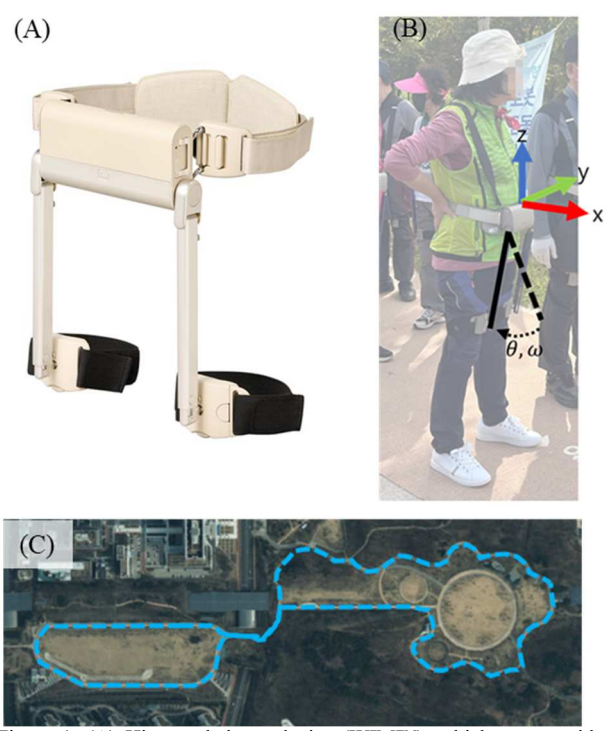

Figure 1. (A) Hip exoskeleton device (WIM™), which can provide assistive or resistive torque on hip joint during walking. (B) A participant wearing the device with the location of an IMU and an encoder visualized. (C) Walk trajectory overlayed by GPS result (blue dash)

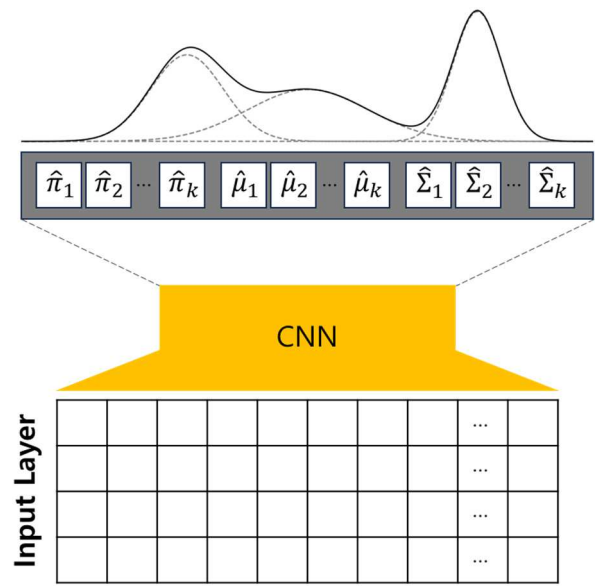

Figure 2. Final shape of the network architecture utilized on this article (CMDN). MDN is put on the top of the CNN architecture.

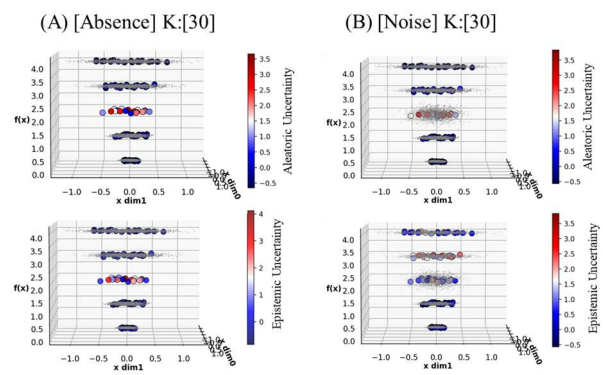

Figure 3. Uncertainty plots on the synthetic dataset. (A) simulation on absence of data, (B) simulation on extra noise

institutional review board of the Korea National Institute for Bioethics Policy (P01-202309-01-032).

B. Deep Neural Network

In this article, the baseline regression performance was measured using the convolution neural network (CNN) structure [5], [17]. However, hyperparameter tuning was performed considering the specifications of the MCU (STM32G4 series [18]) currently used in the device. Applying the method used for EfficientNet [19], modifications were made to the model depth, width and resolution to maintain the model complexity while improving the regression accuracy. In specific, five 1-dimensional convolution layers (depth = 5) were used, with the first layer having 12 channels and a kernel size of 3. In the remaining layers, the number of channels is doubled and then gradually decreased with a kernel size of 5 (Fig. 2). Acceleration in 3 different axis and angular velocity signal magnitude vector were used as input data for 1 second [20], resulting in an input data format of a $[20 \times 4]$ matrix.

C. Mixture Density Network

In addition to the baseline CNN model, MDN attached CNN (CMDN) was used. The same CNN architecture was used, and on the last layer MDN was placed. The MDN is

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employed to decompose a give probability distribution into k Gaussian distributions [15]. Thus, the last layer of the CNN is configured to provide $3 \times k$ outputs. These outputs are further trained into k mixture weights (π), means (μ), and standard deviations (σ). The final output utilizes the mean of the mixture with the highest weight (Fig. 2) In this article, $k = 30$ was used and the entire network is implemented using PyTorch [21].

Two uncertainties of the output based on the input is calculated according to the equations below [15], [16].

$$\mathbf{V}_{k\sim\pi}(\mathbf{E}[\boldsymbol{y}|\boldsymbol{x},k]) = \sum_{k=1}^{K} \pi_k(\boldsymbol{x}) \left\| \mu_k(\boldsymbol{x}) - \sum_{j=1}^{K} \pi_j(\boldsymbol{x})\mu_j(\boldsymbol{x}) \right\|^2 \quad (1)$$

$$\mathbf{E}_{k\sim\pi}[\mathbf{V}(\boldsymbol{y}|\boldsymbol{x},k)] = \sum_{k=1}^{K} \pi_k(\boldsymbol{x})\Sigma_k(\boldsymbol{x}) \quad (2)$$

On the synthesized dataset of input space and output data following $X \in \mathbb{R}^{10000\times20\times2}$, X_i is sampled from a normal distribution with mean 0 and varying standard deviation on the formula $\frac{floor\left(\frac{i}{2000}+1\right)}{3} + 10^{-4}$, $y \in \mathbb{R}^{10000\times1}$, $y_i = floor\left(\frac{i}{2000}\right)$, CMDN was applied and tested. To assess the functionality of the two uncertainties, data for the following two scenarios were synthesized; a) when the training data in range of 4000 to 6000 were removed, b) when random uniform noise with mean 0 and standard deviation 0.5 was added to the training data in range of 4000 to 6000. The test results showed (Fig. 3) that in scenario a) epistemic uncertainty (1) increased and in scenario b) aleatoric uncertainty (2) was increased [15].

D. Model Validation

To assess and compare the performance of the baseline CNN and CMDN, the root mean squared error (RMSE) between the GPS-based speed calculation results is utilized. In this experimental study, twenty elderly individuals actively participated, contributing data while walking on various environment including flat terrain, slopes and stairs. Meanwhile, the torque options of the wearable device were adjusted based on the personal preference. Subsequently, after the collection of training data under the specific torque conditions (torque magnitude = 3 Nm, assist mode), the models were tested under the different torque conditions (torque magnitude = -3 Nm, resist mode). The evaluation of the average RMSE of the test dataset serves as the final performance metric.

To assess uncertainty performance, aleatoric and epistemic uncertainties were compared in randomly sampled data from each of both the level ground walking and stair descent conditions, with 100 data points for each condition. This comparison aimed to provide insights into the variations and distinctions between uncertainties under different scenarios, contributing to a comprehensive understanding of the model's behavior. The validation of the average uncertainty in each condition was conducted through a two-way repeated measures analysis of variance (ANOVA) using MATLAB R2023b, MathWorks.

TABLE I. EVALUATION OF THE MODELS

	CNN	CMDN
RMSE (m/sec)	0.1868	**0.1400**

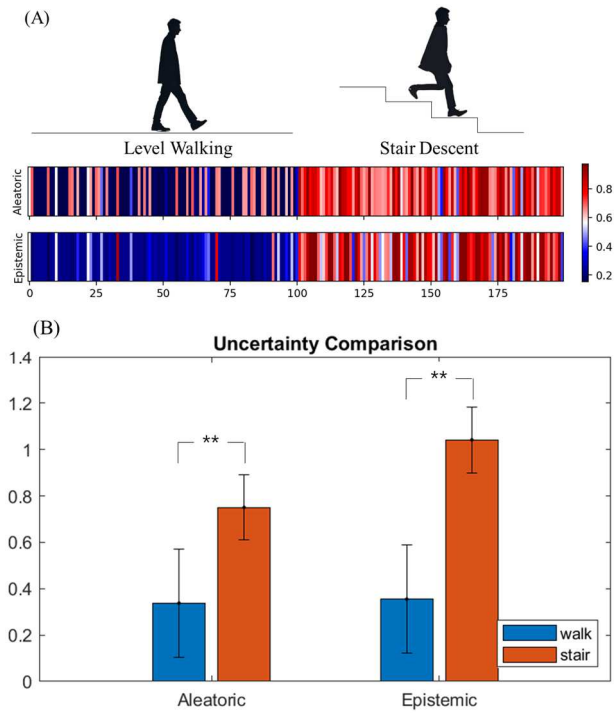

Figure 4. Epistemic and aleatoric uncertainties in walking / stair descent situations. (A) Uncertainty heatmap. (B) Bar plot with ANOVA result. **p_value < 0.001

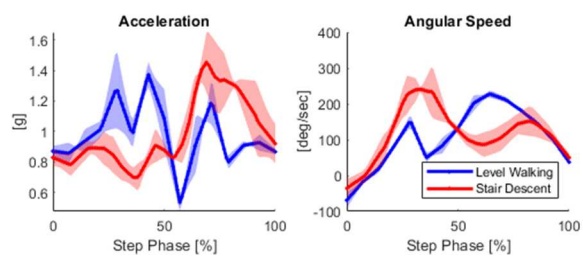

Figure 5. Raw sensor values (acceleration in gravity direction and joint angular speed) during level walking and stair descent

III. RESULTS

A. Accuracy Enhancement with MDN

The RMSE of CMDN is 0.1400 m/sec, which is lower than the baseline CNN's 0.1868 m/sec [TABLE I]. The result underscores the effectiveness of incorporating MDN, especially in scenarios where the test set operates in a different control environment from the training set. Noting that the RMSE result is calculated on the test set of different environments from the training set, this improvement suggests that CMDN not only outperforms the baseline in typical conditions but also demonstrates enhanced robustness across diverse environmental settings.

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B. Uncertainty Estimation in Various Environments

When comparing uncertainties in level walking and stair descent, both aleatoric and epistemic uncertainties increased in the stair descent scenario by 0.4129 ± 0.2659 and 0.6864 ± 0.3122, respectively (p-value < 0.001) (Fig. 4).

When checking the raw sensor values (Fig. 5), both IMU acceleration and hip joint angular speed trajectories vary between level walking and stair descent situation. The observed rise in aleatoric uncertainty in stair descent suggests environmental noise compared to the level walking environment. Similarly, the increase in epistemic uncertainty can be linked to the occurrence of sensor values not covered by the level walking data. These findings show the importance of understanding how uncertainty behaves in various environmental conditions and its potential impact on model performance. Future research should cover exploring a broader range of scenarios to comprehensively characterize the patterns of uncertainty variations.

IV. Discussion and Conclusion

In this study, we explored the integration of Mixture Density Network (MDN) into a CNN model designed to estimate walking speed using the IMU sensor embedded in wearable devices. The results not only showed an enhancement in performance but also highlight the potential of utilizing uncertainty for accurate walking environment estimation. Given the dynamic nature of the control environment in wearable devices, user-specific characteristics, and the variability in walking environments, the demand for a robust and reliable walking speed estimation algorithm becomes crucial. The usage of MDN has significant advantages in these aspects compared to traditional regression methods.

In the current uncertainty estimation process, our focus was solely on utilizing IMU values. Recognizing the obvious changes in sensor values, particularly leg angles, future works should leverage encoder-measured leg angle values for a more detailed uncertainty evaluation. Moreover, uncertainty variance according to broader environmental changes, especially difference between the aspects of aleatoric uncertainty and epistemic uncertainty is needed. The insights gained from those studies will contribute to enhancing the reliability of walking ability estimations.

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